# Improving Robustness in LLM Math Reasoners Under Input Perturbations

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**Abstract**

Large language models (LLMs) exhibit significant sensitivity to surface-level input perturbations, raising the question of whether these models are truly reasoning or exploiting superficial patterns. In this work, we investigate robustness of LLM math reasoners under input perturbations using the GSM-Plus benchmark [2]. We propose a consistency loss training objective that encourages the model's hidden representations to remain stable across selected perturbation pairs where representation alignment is reasonable. We conduct a three-way ablation: (1) fine-tuning on original questions only (SFT-orig), (2) fine-tuning on original and perturbed questions (SFT-both), and (3) adding our consistency loss on top (SFT-cons). We further ablate over target layer ($l^* \in \{8, 18, 23\}$) and gradient flow direction (symmetric vs. anchored). We compare against a base Qwen2.5-Math-1.5B model [3], a supervised fine-tuned baseline, and a GRPO-trained model [4]. Our hidden state analysis identifies layer 23 as the point of maximum divergence between original and perturbed representations. Applying consistency loss at this layer improves robustness on 4 out of 7 perturbation types, with the largest gains on distraction insertion and numerical substitution.

Code is available at `https://github.com/jungsuakim/llm-math-robustness`.

## 1 Introduction

Recent work has demonstrated that LLMs can be highly sensitive to input perturbations that preserve the underlying mathematical problem but alter its surface form [1, 2]. This sensitivity suggests that models may be relying on superficial lexical or structural cues rather than performing genuine mathematical reasoning. Understanding and improving robustness under such perturbations is critical for deploying LLMs in real-world settings where problem phrasings vary widely.

In this work, we study robustness under input perturbations using GSM-Plus [2], a benchmark that provides pairs of original and perturbed input problems, extending GSM8K with eight types of perturbations including digit expansion, numerical substitution, distraction insertion, and problem rephrasing. We focus on the Qwen2.5-Math-1.5B model family and ask: can we improve robustness to semantically equivalent perturbations through targeted fine-tuning and representation-level consistency objectives?

Our main contributions are: (1) an evaluation of Qwen2.5-Math-1.5B under GSM-Plus perturbations across base, SFT, GRPO, and fine-tuned variants; (2) a layerwise hidden state similarity analysis identifying layer 23 as the most perturbation-sensitive layer; (3) a consistency loss training method with ablations over target layer and gradient direction; and (4) a finding that training on perturbed data improves absolute accuracy, while consistency loss at layer 23 reduces the accuracy gap on 4 of 7 perturbation types.

## 2 Related Work

Robustness of LLMs to input perturbations has been studied across a variety of tasks. Ribeiro et al. [5] introduced CheckList, a behavioral testing framework that revealed systematic failures in NLP models under simple transformations. In the mathematical reasoning domain, Patel et al. [6] showed that models trained on GSM8K are brittle to numerical substitutions and template changes.

MATH-Perturb [1] systematically benchmarks frontier LLMs under hard and simple perturbations, finding consistent accuracy degradation on hard perturbations. GSM-Plus [2] similarly constructs eight perturbation types over GSM8K and finds that even strong models can degrade under perturbed problem statements. These benchmarks motivate our focus on robustness to math word-problem perturbations.

Several approaches have been proposed to improve or analyze robustness. Representation regularization methods such as contrastive learning [7] and consistency training [8] encourage stable representations across related inputs. Interpretability methods also motivate our layer-targeted approach: the logit lens [9] enables inspection of intermediate predictions across layers, while ROME [10] suggests that specific computations can be localized to particular transformer layers. Our work combines these ideas by applying a cosine-similarity consistency loss directly to hidden states at the layer where original and perturbed inputs diverge most.

Supervised fine-tuning on mathematical reasoning data has been shown to improve model accuracy in systems such as Qwen2.5-Math [3], while GRPO-style training has been used to further improve mathematical reasoning performance [4]. We use SFT- and GRPO-trained checkpoints as baselines for comparison.

## 3 Method

### 3.1 Problem Setup

Given a math problem $x$ and a perturbed version $x'$, we study whether model behavior remains appropriate under changes to the problem statement. GSM-Plus includes perturbations with different effects: some preserve the correct answer, such as distraction insertion and problem understanding, while others modify numerical values or operations and therefore change the correct answer. Because of this, we distinguish between accuracy and robustness rather than treating answer consistency as universally desirable.

Our primary robustness-related metric is the *accuracy gap*, defined as original accuracy minus perturbed accuracy. A smaller gap indicates that model performance is less sensitive to perturbation, though it does not fully capture all aspects of robustness. We also report *answer consistency*, defined as the fraction of examples where the model gives the same final answer to $x$ and $x'$, and interpret it cautiously.

### 3.2 Perturbation Types

We use GSM-Plus [2] which provides eight perturbation types over GSM8K. We exclude *critical thinking* perturbations from both training and evaluation: these perturbations fundamentally change the problem's answer (perturbed accuracy drops to near zero), making consistency impossible by design and providing no useful training signal. We additionally exclude *reversing operation* from training only, since it also changes the correct answer, but include it in evaluation to measure generalization. The six perturbation types used for training are: digit expansion, integer-decimal-fraction conversion, numerical substitution, distraction insertion, problem understanding, and adding operation.

### 3.3 Hidden State Similarity Analysis

To identify the layer most sensitive to perturbations, we compute the cosine similarity between hidden states of original and perturbed inputs at each layer of Qwen2.5-Math-1.5B:

$$\text{sim}_l = \text{cosine}\big(\mathbf{h}_l(x_{\text{orig}}),\ \mathbf{h}_l(x_{\text{pert}})\big)$$

where $\mathbf{h}_l(x)$ is the hidden state at the last token position at layer $l$, averaged over 20 GSM-Plus pairs. The base model exhibits minimum similarity at layer 23 (0.9081), with a steady decline from layer 5 onward. The SFT model shows a similar profile, also reaching minimum similarity at layer 23, while GRPO maintains higher similarity throughout and does not show a clear sensitive layer. We use $l^* = 23$ as the primary target motivated by this analysis, and also ablate $l^* \in \{8, 18\}$.

### 3.4 Consistency Loss

We augment the standard cross-entropy loss with a consistency regularization term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \big(1 - \text{cosine}(\mathbf{h}_{l^*}(x_{\text{orig}}),\ \mathbf{h}_{l^*}(x_{\text{pert}}))\big)$$

where $\lambda = 0.01$. We study two gradient flow variants: (1) **symmetric** – gradients flow through both $\mathbf{h}_{l^*}(x_{\text{orig}})$ and $\mathbf{h}_{l^*}(x_{\text{pert}})$, pulling both toward each other; (2) **anchored** – $\mathbf{h}_{l^*}(x_{\text{orig}})$ is detached, so only the perturbed representation moves toward the original. We train on original and perturbed question-solution pairs jointly, averaging the CE losses across both.

### 3.5 Training Setup

We fine-tune Qwen2.5-Math-1.5B with learning rate $5 \times 10^{-6}$, batch size 1, gradient clipping 1.0, for 2 epochs. Training data uses GSM-Plus test indices 400–10552 filtered to 6 valid perturbation types (≈7,500 pairs). Evaluation uses indices 0–400 with 50 examples per type (350 total), excluding critical thinking. We compare:

1. **SFT-orig**: CE loss on original questions only.
2. **SFT-both**: CE loss on both original and perturbed questions.
3. **SFT-cons**: CE loss on both questions + consistency loss, with ablations over $l^* \in \{8, 18, 23\}$ and symmetric vs. anchored gradient flow.

## 4 Experiments

### 4.1 Baselines

We compare against three baselines on the same 350 GSM-Plus evaluation examples:

1. **Base**: Qwen2.5-Math-1.5B with no fine-tuning.
2. **SFT (a3)**: Qwen2.5-Math-1.5B fully fine-tuned on MATH (lr=2e-5, bs=16, grad accum=4, 2 epochs).
3. **GRPO (a3)**: Qwen2.5-Math-1.5B fine-tuned with group relative policy optimization on MATH [4].

### 4.2 Main Results

| Model | Orig Acc | Pert Acc | Gap | Consistency |
|---|---|---|---|---|
| Base | 0.620 | 0.529 | 0.091 | 0.189 |
| SFT (a3) | 0.451 | 0.366 | 0.086 | 0.126 |
| GRPO (a3) | 0.743 | 0.666 | 0.077 | 0.263 |
| SFT-orig (ours) | 0.623 | 0.566 | 0.057 | 0.211 |
| SFT-both (ours) | 0.709 | 0.606 | 0.103 | 0.254 |
| SFT-cons $l^* = 8$ sym. | 0.689 | 0.597 | 0.091 | 0.243 |
| SFT-cons $l^* = 8$ anchor | 0.720 | 0.594 | 0.126 | 0.243 |
| SFT-cons $l^* = 18$ sym. | 0.686 | 0.600 | 0.086 | 0.257 |
| SFT-cons $l^* = 23$ sym. | 0.671 | 0.589 | **0.083** | 0.234 |
| SFT-cons $l^* = 23$ anchor | 0.703 | 0.600 | 0.103 | 0.251 |

Table 1: Main results on GSM-Plus (350 examples, 50 per perturbation type, critical thinking excluded). Note that given the evaluation set size, differences smaller than 5 percentage points should be interpreted cautiously.

### 4.3 Per-Perturbation-Type Analysis

| Perturbation Type | Base | SFT-orig | SFT-both | SFT-cons $l^* = 23$ |
|---|---|---|---|---|
| distraction insertion | 0.20 | 0.18 | 0.14 | 0.10 |
| adding operation | 0.14 | 0.18 | 0.22 | 0.26 |
| reversing operation | 0.12 | 0.10 | 0.08 | 0.10 |
| int.-dec.-frac. conversion | 0.12 | 0.04 | 0.06 | 0.06 |
| numerical substitution | 0.08 | $-0.02$ | 0.06 | 0.00 |
| digit expansion | 0.02 | $-0.02$ | 0.08 | 0.04 |
| problem understanding | $-0.04$ | $-0.06$ | 0.08 | 0.02 |

Table 2: Accuracy gap per perturbation type (Orig Acc $-$ Pert Acc. Negative gap = model performs better on perturbed than original (50 examples per type). SFT-cons $l^* = 23$ improves over base on 4 of 7 types: distraction insertion, int.-dec.-frac. conversion, numerical substitution, and problem understanding.

Additionally, qualitative inspection reveals two dominant failure patterns: distraction incorporation and format sensitivity (see Appendix A).

# 5 Analysis and Discussion

## 5.1 Hidden State Analysis and Layer Selection

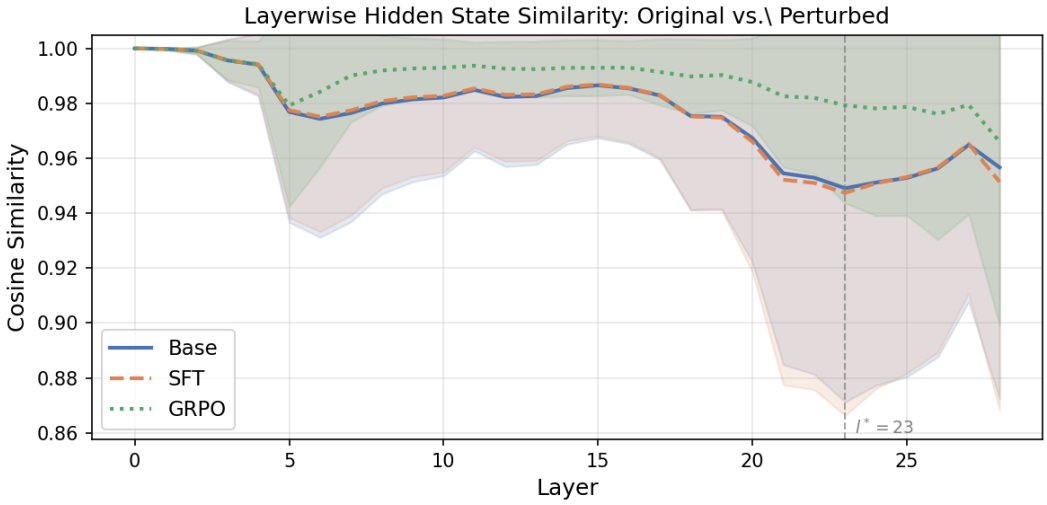


Figure 1: Layerwise cosine similarity between hidden states of original and perturbed inputs, averaged over 20 GSM-Plus pairs (indices 400+). Shaded regions show $\pm 1$ std. Base and SFT show nearly identical profiles with minimum at layer 23, while GRPO maintains higher similarity throughout. The vertical dashed line marks $l^* = 23$.

The base model's similarity profile drops steadily from layer 5, reaching a minimum at layer 23 (0.9081). Base and SFT show nearly identical profiles with minimum similarity at layer 23, while GRPO maintains consistently higher similarity throughout with no clear sensitive layer, consistent with its stronger overall robustness. This motivates our primary choice of $l^* = 23$ for consistency loss targeting.

To further validate this choice, we apply the logit lens [9] to compute KL divergence between the token prediction distributions of original and perturbed inputs at each layer. The mean peak KL divergence occurs at layer 22.7, consistent with the cosine similarity minimum at layer 23. KL divergence at $l^* = 23$ (0.022) is also about $4\times$ higher than at $l^* = 8$ (0.006), suggesting that divergence near layer 23 has a larger effect on the model's intermediate predictions.

## 5.2 Does Training on Perturbed Data Help?

## 5.3 Effect of Consistency Loss and Layer Choice

Among the consistency-loss variants, $l^* = 23$ symmetric achieves the lowest gap (0.083). Examining per-type results in Table 2, SFT-cons $l^* = 23$ improves over the base model on **4 of 7 perturbation types**: distraction insertion (0.20→0.10), integer-decimal-fraction conversion (0.12→0.06), numerical substitution (0.08→0.00), and problem understanding (−0.04→0.02). One failure case to note is adding operation, where the gap actually increases from 0.14 to 0.26.

Notably, SFT-cons $l^* = 23$ removes the negative gap on *problem understanding*, meaning the model no longer does better on the perturbed version than the original. This is consistent with our hidden state analysis: layer 23 is where the base model's representations diverge most, and aligning them there specifically improves robustness to semantic rephrasing.

In contrast, $l^* = 8$ symmetric matches the base gap exactly (0.091) despite higher original accuracy (+0.069 vs. +0.051 for $l^* = 23$). This suggests a tradeoff between the two layers: earlier-layer alignment may improve original-question accuracy more, while later-layer alignment better reduces the original–perturbed accuracy gap.

### 5.4 Symmetric vs. Anchored Gradient Flow

The anchored variant (detach original, only perturbed moves) consistently raises original accuracy but worsens the gap: $l^* = 8$ anchor (0.720 orig, gap 0.126) vs. symmetric (0.689 orig, gap 0.091). Similarly for $l^* = 23$: anchor (0.703 orig, gap 0.103) vs. symmetric (0.671 orig, gap 0.083). The anchor approach treats the original as a fixed target, improving the model's original-question performance but not its ability to generalize to perturbations. Symmetric gradient flow produces smaller gaps, suggesting it is better suited for reducing perturbation sensitivity.

### 5.5 GRPO as a Strong Baseline

GRPO (0.743 orig, 0.666 pert, gap 0.077, consistency 0.263) achieves the best overall performance. Its gap (0.077) is smaller than base (0.091) despite much higher accuracy, suggesting RL-based training may confer implicit robustness benefits, possibly through more structured step-by-step reasoning [4].

### 5.6 Limitations

Our study has several limitations. First, we train and evaluate on GSM-Plus only, which covers arithmetic reasoning but not more complex mathematical domains. Second, $\lambda = 0.01$ was fixed; stronger regularization might improve gap reduction. Third, the evaluation set is small (350 examples, 50 per type) — given this size, the standard error of a proportion is approximately $\pm 5\%$, meaning differences smaller than 5 percentage points may not be statistically significant. The per-type results (50 examples each) are especially noisy, and differences below 10 percentage points should be interpreted cautiously. Fourth, we only study one model family and size.

## 6 Conclusion

We investigated two approaches to improving robustness of LLM math reasoners under input perturbations. First, fine-tuning on perturbed examples (SFT-both) substantially improves absolute accuracy on both original and perturbed inputs, showing that exposure to diverse phrasings during training is a simple and effective robustness intervention. Second, adding a consistency loss that encourages aligned hidden representations partially mitigates the increased gap introduced by training on both original and perturbed examples, especially when applied at layer 23. Our hidden state analysis identifies layer 23 as the point of maximum divergence between original and perturbed representations in the base model, and applying consistency loss there improves robustness on 4 of 7 perturbation types, with the largest gains on distraction insertion (gap 0.20→0.10) and numerical substitution (gap 0.08→0.00). Types involving structural changes to the problem (adding operation, digit expansion) do not benefit, suggesting the consistency loss is most effective for surface-level reformulations.

Future work should explore a larger evaluation set, different models, larger $\lambda$, multi-layer consistency objectives, combining consistency loss with GRPO training, and extending the analysis to harder domains such as MATH-Perturb [1].

## References

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# A Qualitative Failure Mode Analysis

We identify two dominant failure patterns from the base model's outputs:

**Distraction incorporation**: On distraction insertion examples, the model sometimes incorrectly incorporates irrelevant information into its reasoning. For example, given "A robe takes 2 bolts of blue fiber and half that much white fiber. [If each bolt costs $5 but there's a $2 discount today,] how many bolts does it take?", the base model predicts 2 instead of 3 — incorrectly applying the pricing discount to reduce the bolt count. The answer should be independent of price.

**Format sensitivity**: On integer-decimal-fraction conversion, mathematically equivalent reformulations cause failures. For example, "increased the value by 150%" and "increased the value by 1.5 times" are mathematically identical, but the base model gives different answers for each phrasing, suggesting reliance on surface linguistic patterns rather than the underlying mathematical relationship.

These failure modes are consistent with our motivation for the consistency loss: by encouraging aligned representations at layer $l^* = 23$, the model is pushed toward representations less sensitive to these surface variations.